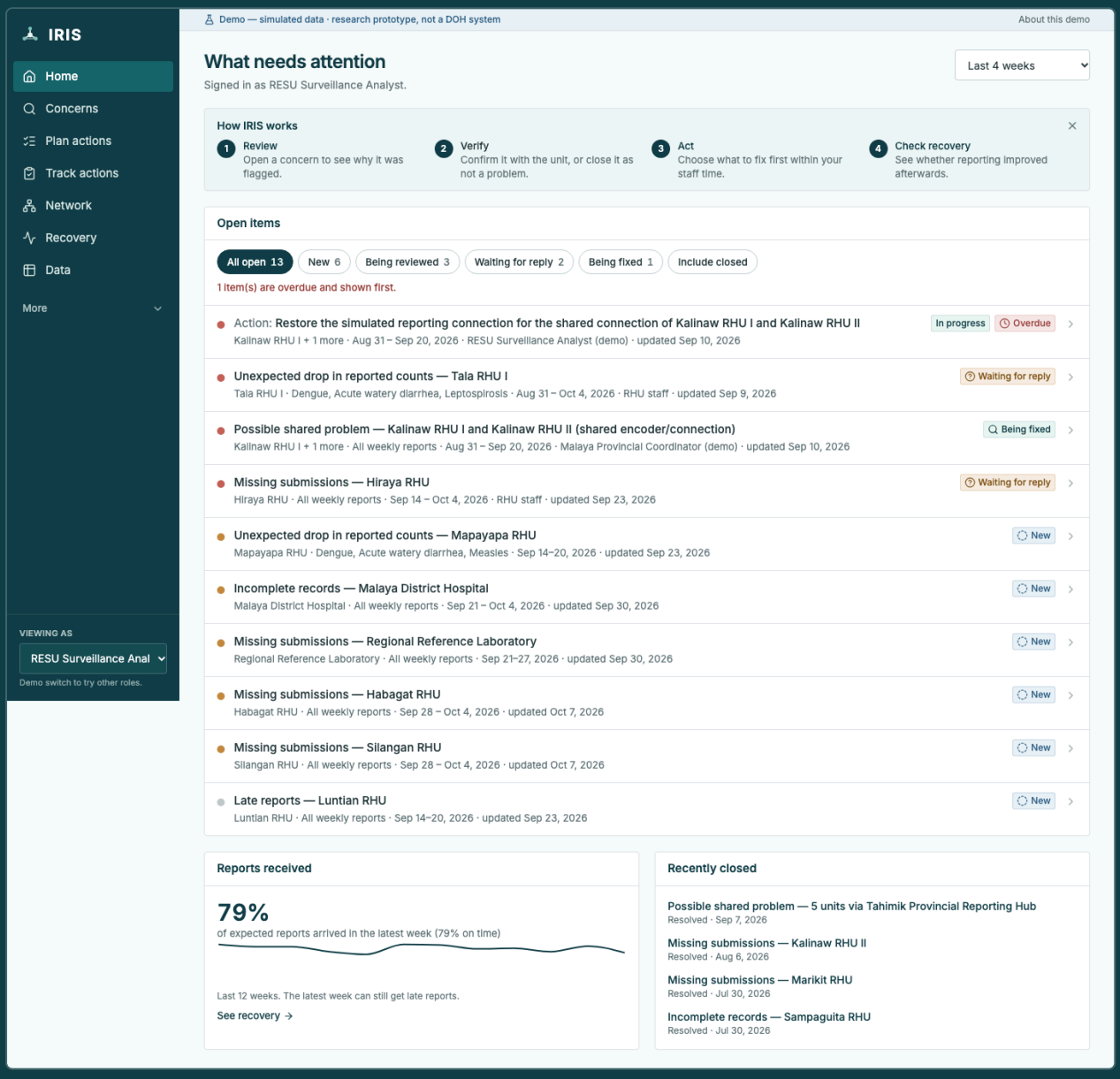
IRIS

User Guide

Integrated Reporting Intervention System — research prototype

> See the connections. Restore what matters.



Version 0.1 · October 2026

All data shown is simulated. Not an official DOH system.

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1. What IRIS does

IRIS helps the people who manage disease-reporting networks notice when reports stop arriving, find out why, decide what to fix first, and check whether reporting recovered.

It is designed for Regional Epidemiology and Surveillance Units (RESUs), provincial coordinators, Rural Health Unit (RHU) staff and data managers. IRIS compares the reports expected from each unit with those that arrived, and raises a **concern** when something looks unusual: missing or late reports, blank forms, or a sudden drop in case counts.

1. **Review**
   Open a concern and see why it was flagged.
2. **Verify**
   Confirm it with the unit, or close it as not a problem.
3. **Act**
   Choose what to fix first within your staff time.
4. **Check recovery**
   See whether reporting improved afterwards.

> **Important:** IRIS is a research prototype. It is not an officially deployed Department of Health system and is not endorsed by any government agency. Every province, municipality, facility, person and relationship in this guide is **simulated**, and no patient information is used.

Key ideas

- **IRIS never acts on its own.** It flags and suggests; people decide. Confirming, closing or approving anything needs a written reason, and every change is recorded.
- **Facts and guesses are kept apart.** Each concern separates what the data shows from possible explanations that still need checking.
- **An unusual number is not proof.** A drop in case counts might be under-reporting, a data-entry slip or a genuine decline. IRIS says so.

2. Getting around

The menu on the left lists the main pages. Less frequently used pages are under **More**. On a phone the menu opens from the button at the top right (see section 14).

Page	Question it answers
Home	What needs attention right now?
Concerns	Which reporting problems were found, and why?
Plan actions	With the staff time we have, what should we fix first?
Track actions	What happened after we acted?
Network	How are reporting units connected? Could problems be linked?
Recovery	Is reporting getting better?
Data	What do the records actually show, and what is missing?
More → **Model evaluation**	How well does IRIS spot real problems?
More → **Data sources**	Where did the data come from, and can it be trusted?
More → **Settings**, **Help**	Display options, accessibility, and definitions.

Roles

What you see depends on your role. In this demo you can switch roles with the **Viewing as** selector at the bottom of the menu. That switch exists only for trying the demo; it is not real sign-in security.

Role	Demo users	Typical work
RHU reporting staff	Hiraya RHU Encoder, Kalinaw RHU I Nurse	Send missing reports, answer questions about their unit, see actions planned for them.
Provincial coordinator	Malaya, Masagana and Tahimik Provincial Coordinators	Review concerns in their province, ask units questions, plan and track actions.
RESU analyst	RESU Surveillance Analyst, RESU Epidemiologist	Everything: region-wide review, checking for new concerns, approving plans, model evaluation.
Administrator / data manager	Data Manager	Import and validate data files, edit reporting-unit details, review the activity log.

A thin blue line at the top of every page reminds you that the data is simulated. Select “About this demo” to read the limitations.

3. Home: what needs attention

Home is the starting point. It lists everything open that you are responsible for, with the most urgent items first: overdue items, then high priority.

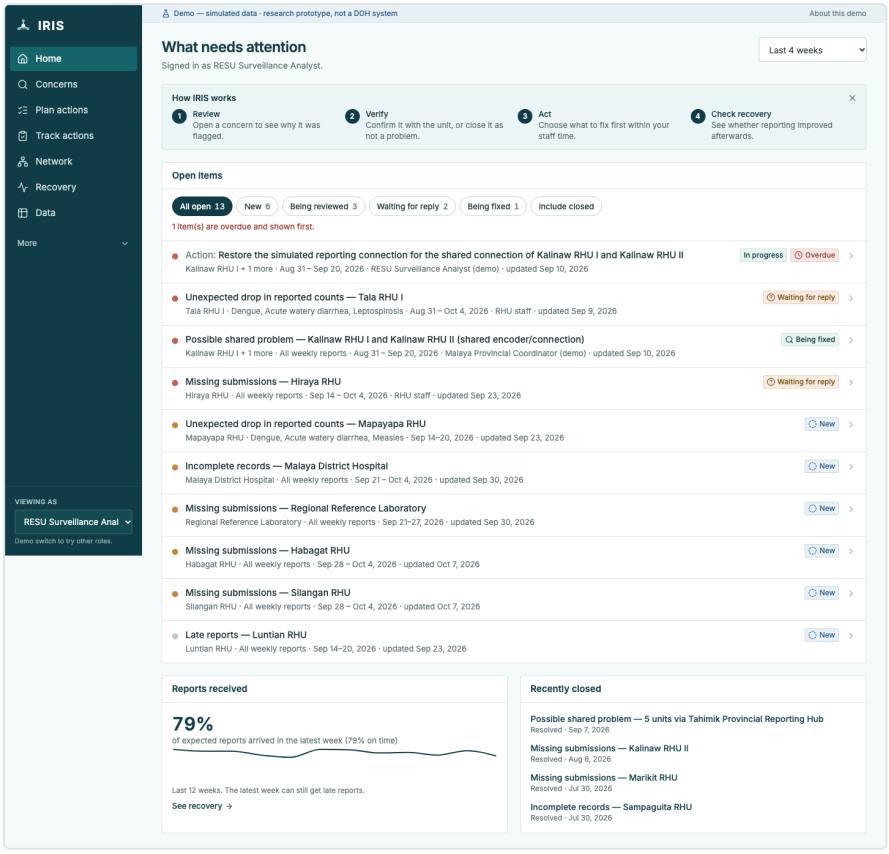


Figure 1. Home for a RESU analyst: the short “How IRIS works” guide, the list of open items, and how many expected reports arrived.

- **How IRIS works**: a four-step reminder. Close it with the ✕ once you no longer need it.
- **Status buttons** (All open, New, Being reviewed, Waiting for reply, Being fixed): show only items in that state. The number is how many there are.
- **Each row** shows a coloured priority dot (red high, amber medium, grey low), the problem, the unit, the reporting weeks and who is handling it. Select a row to open it.
- **Rows starting with “Action:”** are approved fixes in progress, not new problems.
- **Reports received** shows the share of expected reports that arrived in the latest week, with a 12-week trend.
- Use the period selector at the top right (latest week, 4 weeks, 13 weeks, all weeks) to look further back.

> **For RHU staff:** Home shows **Your to-do list** with each missing report and any question from your coordinator. See section 14.

4. Concerns

A **concern** is a possible reporting problem that IRIS found in the data. The Concerns page lists them all.

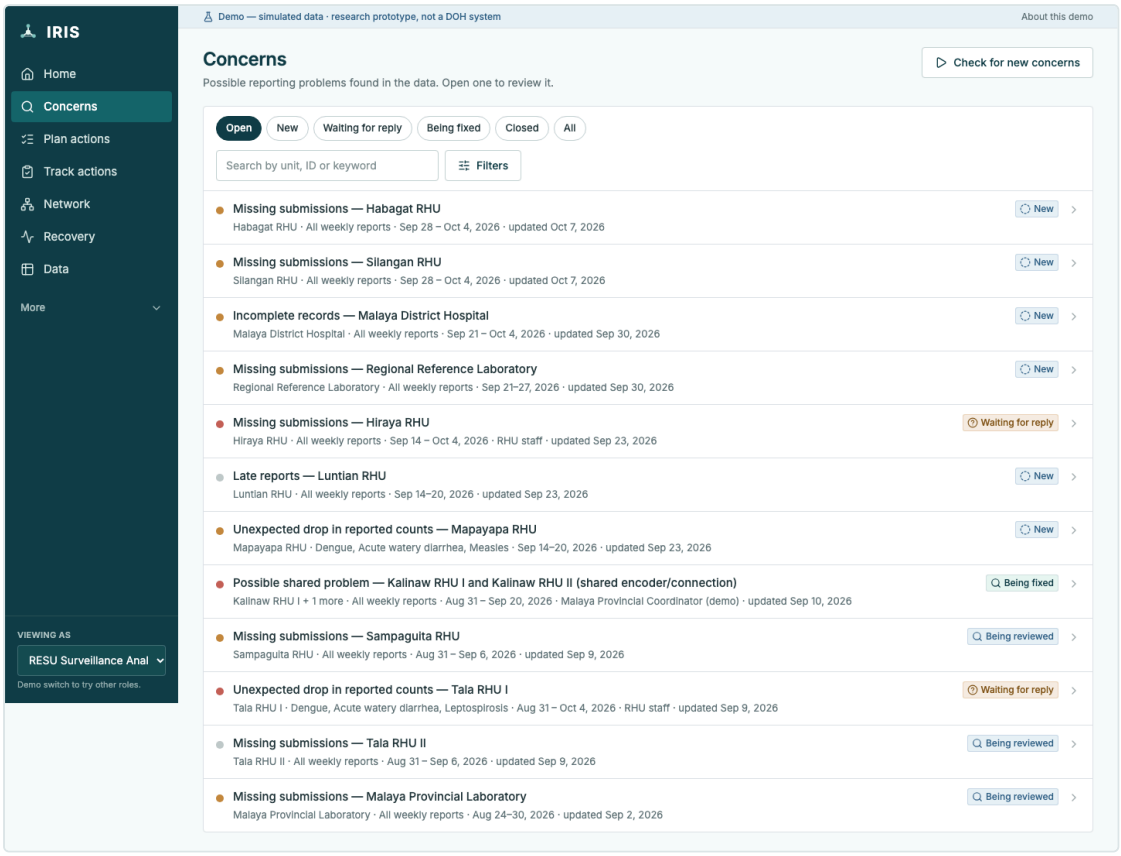


Figure 2. The Concerns list with status buttons, search and the Filters button.

- Use the **status buttons** to switch between open, new, waiting, being fixed, closed or all concerns.
- Type in **Search** to find a unit, a concern ID (for example `C-0010`) or a word.
- Select **Filters** to narrow by type, priority or province.
- Analysts can select **Check for new concerns** to scan the data again. Existing concerns are never changed or duplicated.

Types of concern

Type	Meaning
Missing submission	Expected reports did not arrive.
Late report	Reports arrived after the due date (weekly reports are due Tuesday 5 pm after the week).
Incomplete record	Reports arrived with blank fields or an incomplete form.
Unexpected drop in counts	Reports arrived, but case counts were far below the unit's recent weeks.
Possible shared problem	Several *connected* units had problems at the same time, which may point to a common cause such as a shared hub or internet connection.

5. Working on a concern

Opening a concern shows a progress bar with four steps (**Review → Verify → Act → Close**), a sentence saying what to do **next**, and one main button for that step.

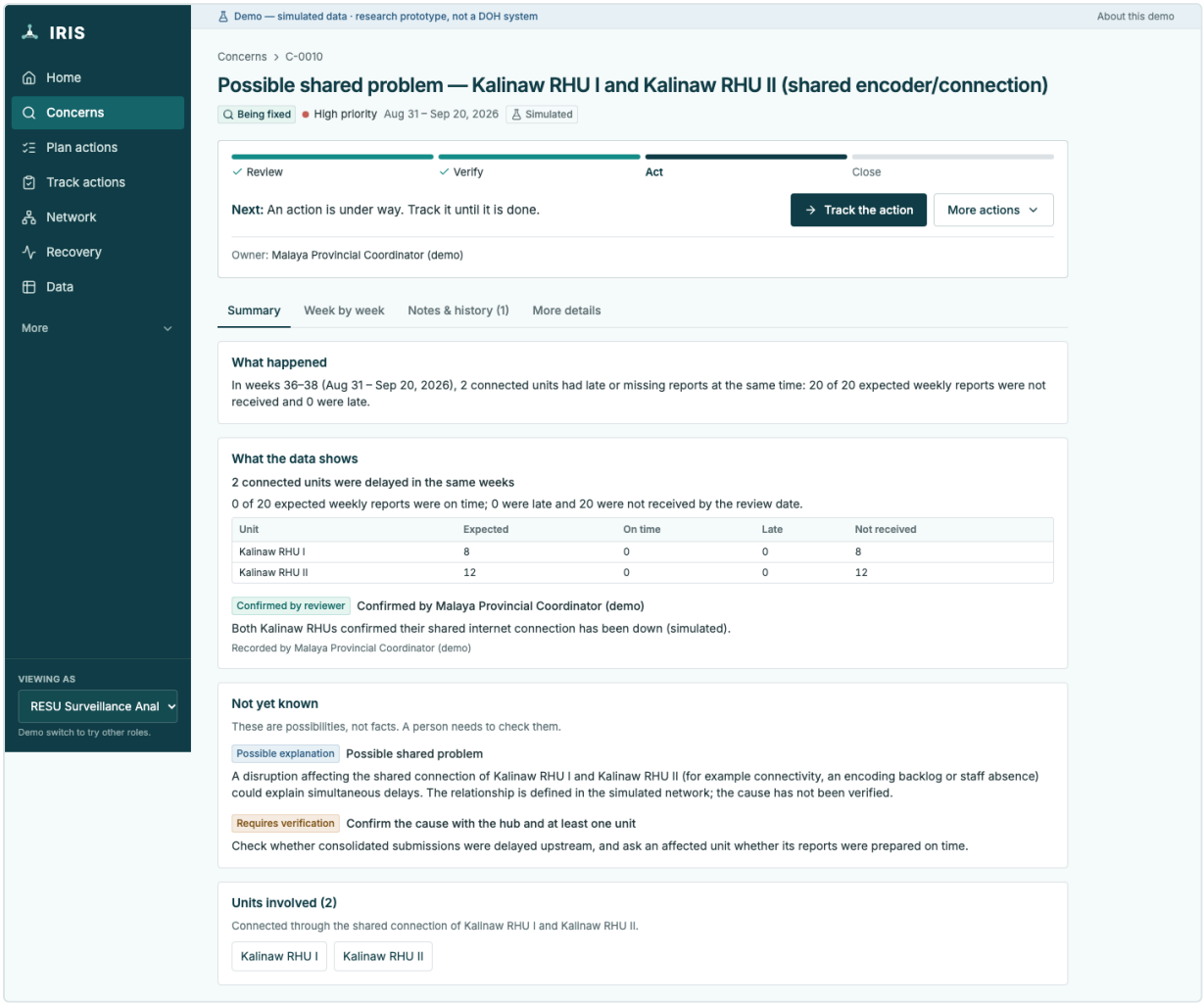


Figure 3. A confirmed shared problem. The next step is to track the action already approved for it.

The page has four tabs:

- **Summary**: *What happened*, *What the data shows* (facts from the records) and *Not yet known* (possible explanations that a person still needs to check), plus the units involved.
- **Week by week**: a simple grid of each unit's reporting per week (Figure 5).
- **Notes & history**: add notes and see every change, who made it and when.
- **More details**: how the concern was detected, reasons recorded, actions taken and related concerns.

Taking action

The main button changes with the step: **Start review**, then **Confirm it's a real problem**, then **Plan an action** or **Track the action**. All other options are in **More actions**.

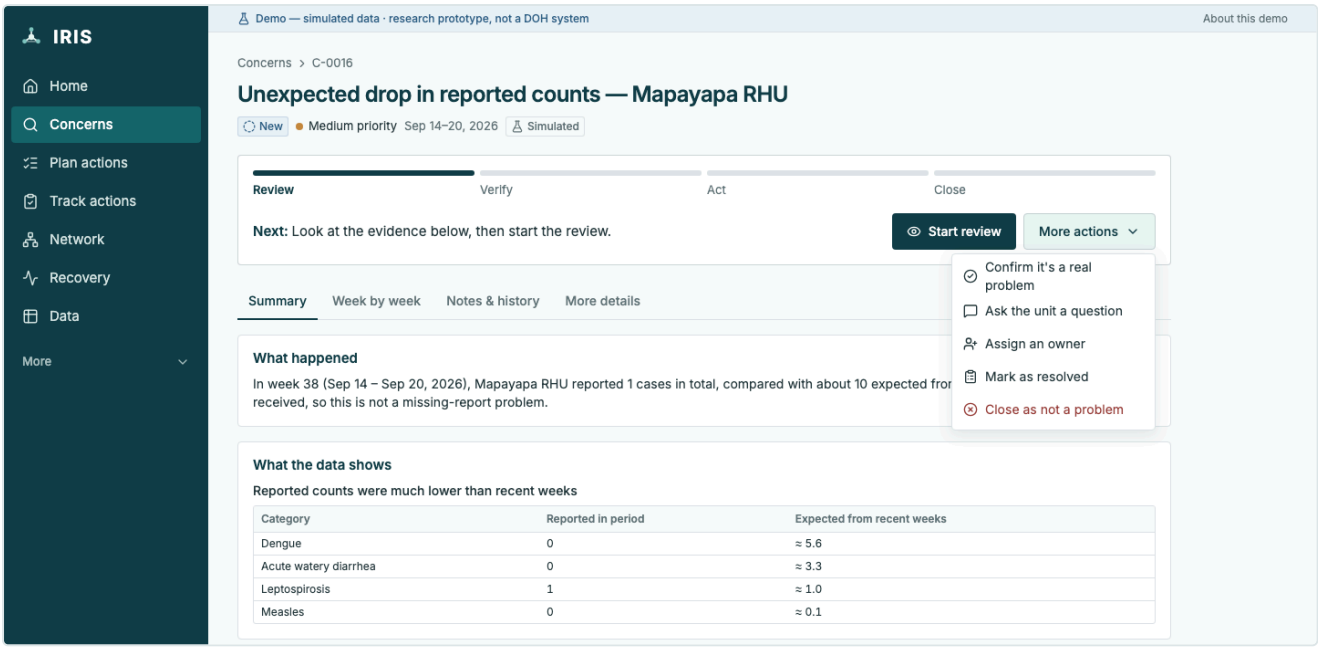


Figure 4. The More actions menu on a new concern. The table under "What the data shows" compares reported counts with recent weeks.

Action	What it does
Start review	Tells others you are looking at it.
Ask the unit a question	Sends a question to the reporting unit; it appears in their to-do list. Status becomes Waiting for reply.
Confirm it's a real problem	Records that the problem was verified. **A short reason is required.**
Assign an owner	Makes one person responsible, with an optional due date.
Mark as resolved	Closes it with a summary of what fixed it. **Required.**
Close as not a problem	Closes a false alarm. **A reason is required.** It can be reopened later.

Statuses

New not looked at yet · Being reviewed someone is checking · Waiting for reply a question was sent to the unit · Being fixed confirmed or assigned · Resolved closed with a summary · Not a problem closed as a false alarm.

Week by week

The **Week by week** tab shows each affected unit as a row and each week as a column. Shaded column headings mark the weeks in this concern.

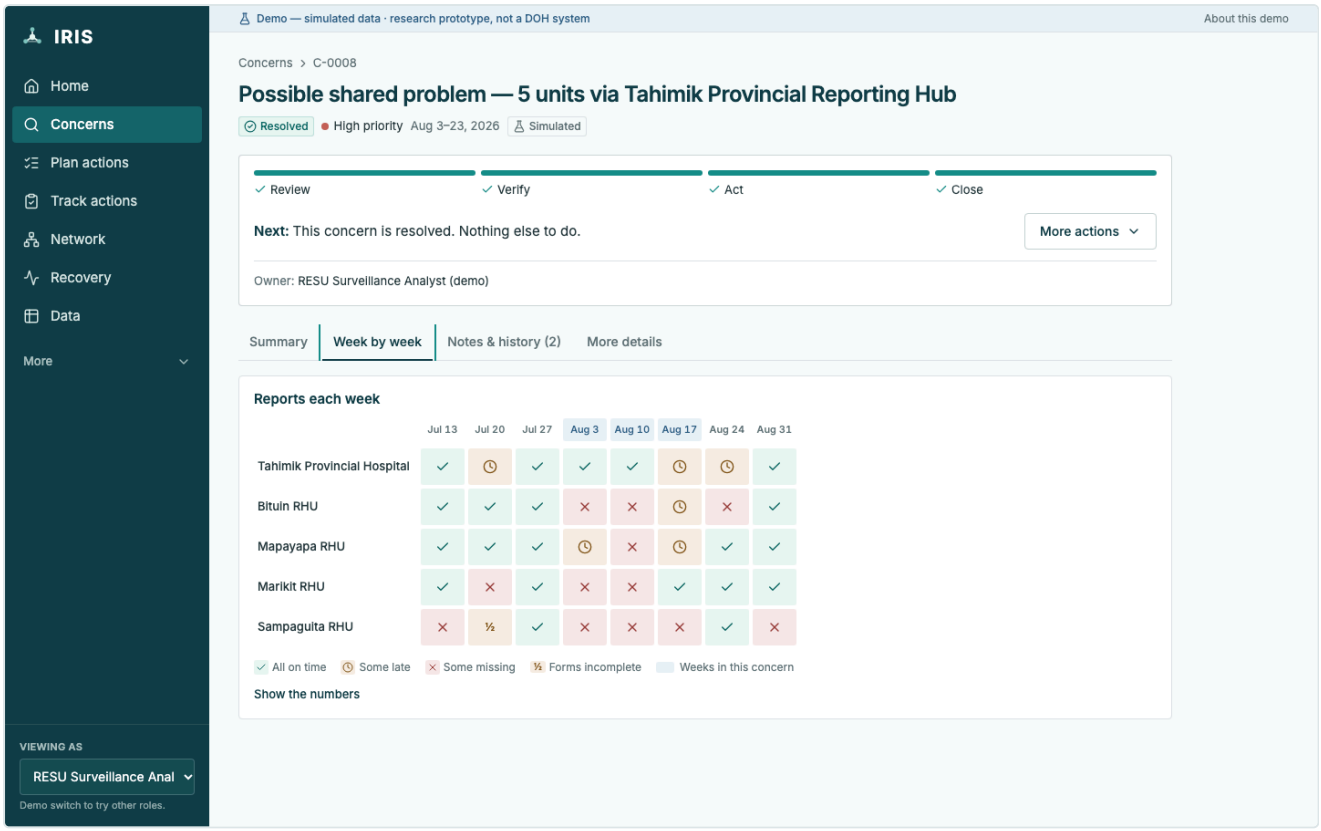


Figure 5. A hub outage: five units reporting through the same hub had missing (×) or late (clock) reports in the same three weeks.

Symbol	Meaning
✓ (green)	All expected reports arrived on time.
Clock (amber)	Some reports were late.
× (red)	Some reports did not arrive.
½ (amber)	Reports arrived but forms were incomplete.

For a drop in counts, the grid shows the number of cases each week instead; red cells are less than half of what was expected. Select **Show the numbers** for the exact counts.

> **Simulation answer key:** under *More details*, the demo can show what was really injected into the simulated data, so you can check your judgement. Turn it off in Settings.

6. Plan actions

When several problems compete for limited staff time, this page suggests which corrective actions to do first.

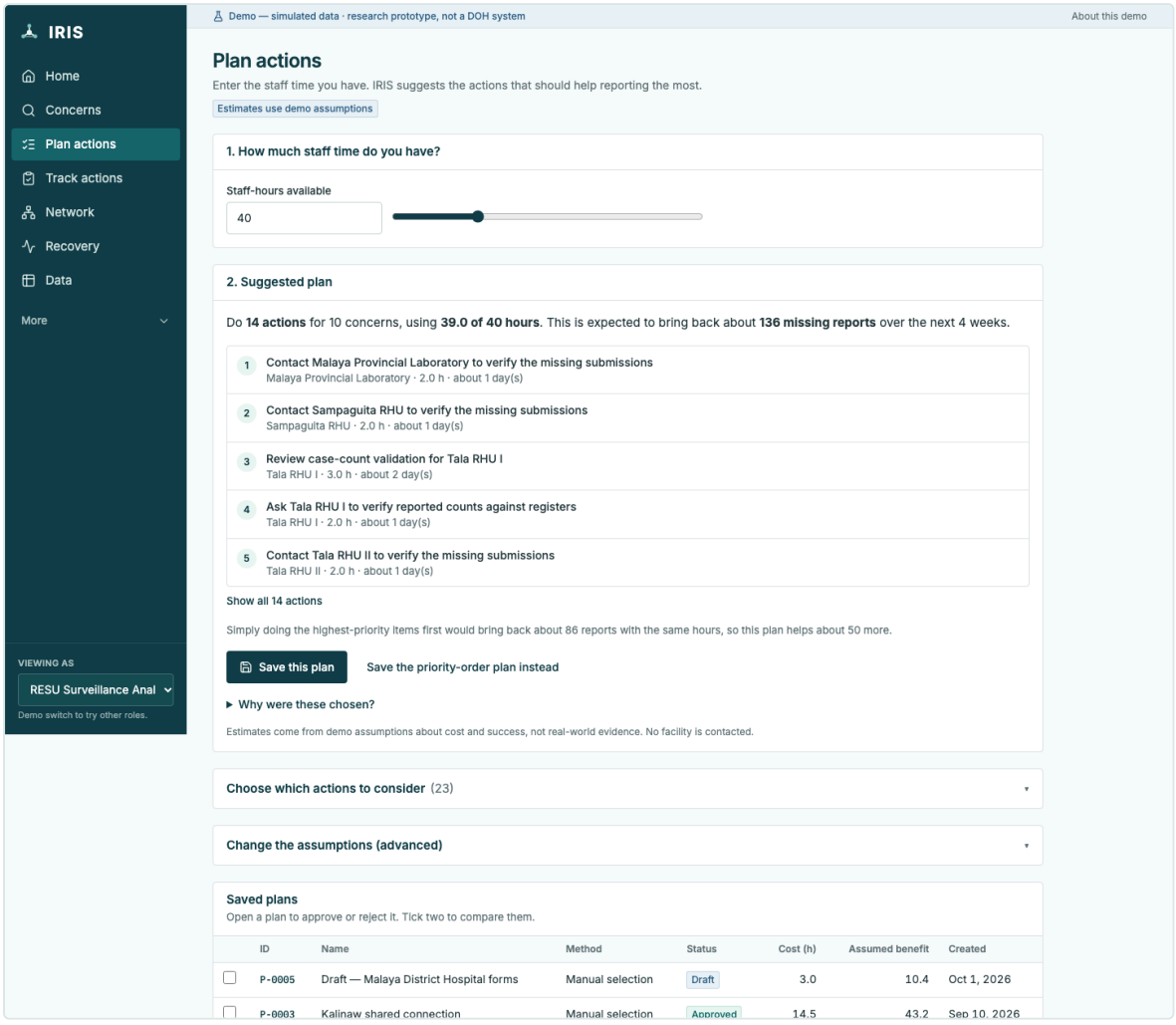


Figure 6. With 40 staff-hours, IRIS suggests a plan and compares it with simply doing the highest-priority items first.

1. Enter the **staff-hours** you have, or drag the slider.
2. Read the **suggested plan**. It updates by itself and says how many actions it includes, the hours used, and roughly how many missing reports it should bring back over the next few weeks.
3. Select **Save this plan**. It is saved as a `Draft`.
4. A coordinator or analyst opens it under **Saved plans** and selects **Approve plan** (or **Reject plan** with a reason). Approval creates the actions in Track actions.

Choose which actions to consider lets you leave actions out. **Change the assumptions** lets advanced users adjust estimated hours and chances of success.

> The estimates come from demonstration assumptions, not evidence of real-world effectiveness. Approving a plan does not contact any facility.

7. Track actions

Approved actions appear here until they are closed. Each one moves through **Planned → Approved → Done → Outcome**.

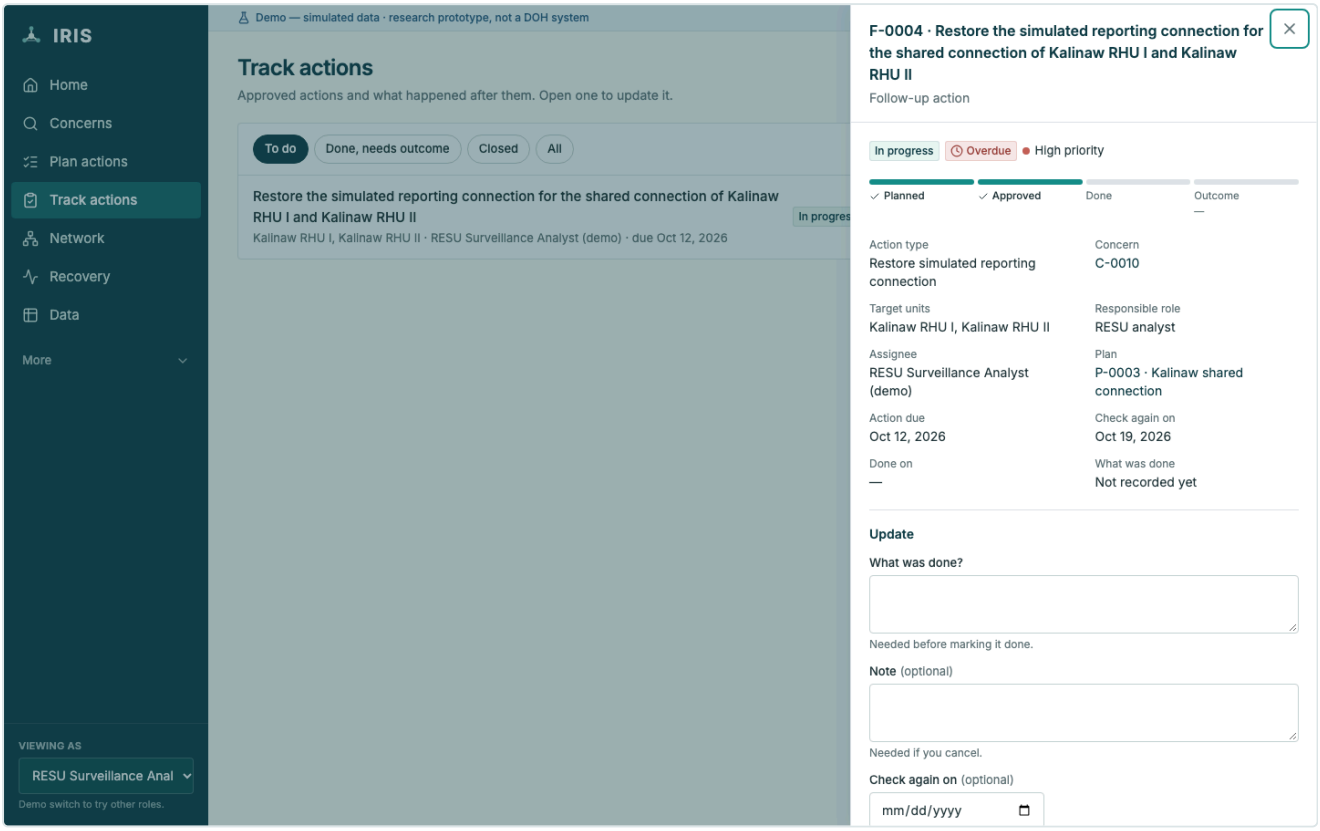


Figure 7. An overdue action opened in the side panel, with its progress and the update form.

- **Start** when work begins.
- **Mark as done**: first describe *what was done*.
- Add a **result**: `Not checked yet`, `Improvement seen, not confirmed`, or `Confirmed by reviewer` (coordinators and analysts only).
- **Close** once a result is recorded. **Cancel action** needs a reason.

Reporting can improve for reasons unrelated to an action, so a result records what was seen, not proof that the action caused it.

8. Network

The Network page shows how reporting units are connected, which helps you judge whether problems in several places might share a cause.

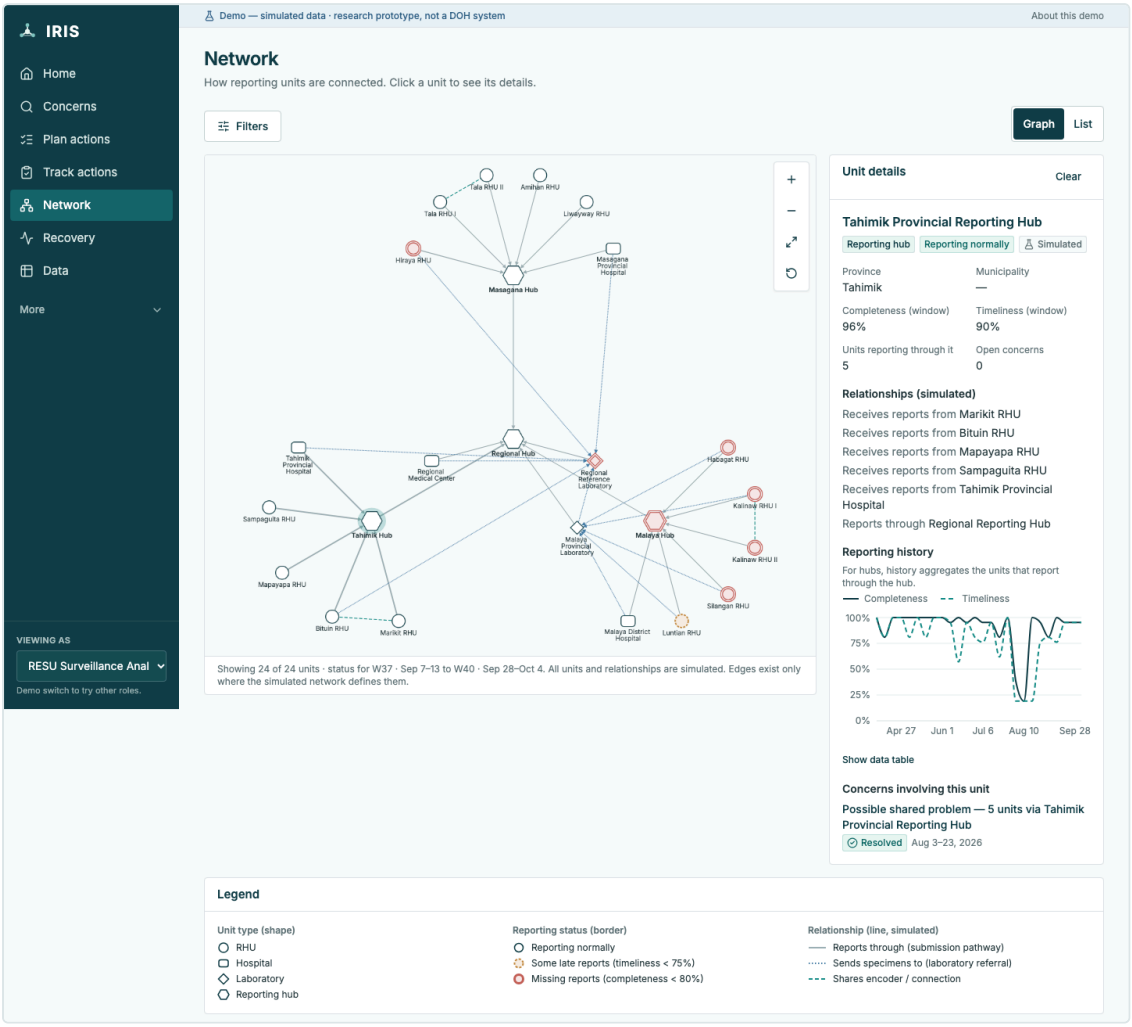


Figure 8. The simulated network with the Tahimik reporting hub selected. The panel shows its connections, reporting history and concerns.

- **Shapes** show the kind of unit: circle = RHU, rounded square = hospital, diamond = laboratory, hexagon = reporting hub.
- **Borders** show recent reporting: plain = normal, dashed amber = some late, thick red = missing reports.
- **Lines**: solid = reports through, dotted = sends specimens to, dashed = shares an encoder or internet connection.
- Click a unit to see its details. Use + / − / fit / reset to move around, or switch to **List** for a table.
- **Filters** limit by province, unit type, weeks or status.

9. Recovery

Recovery answers “is reporting getting better, and did our actions help?”

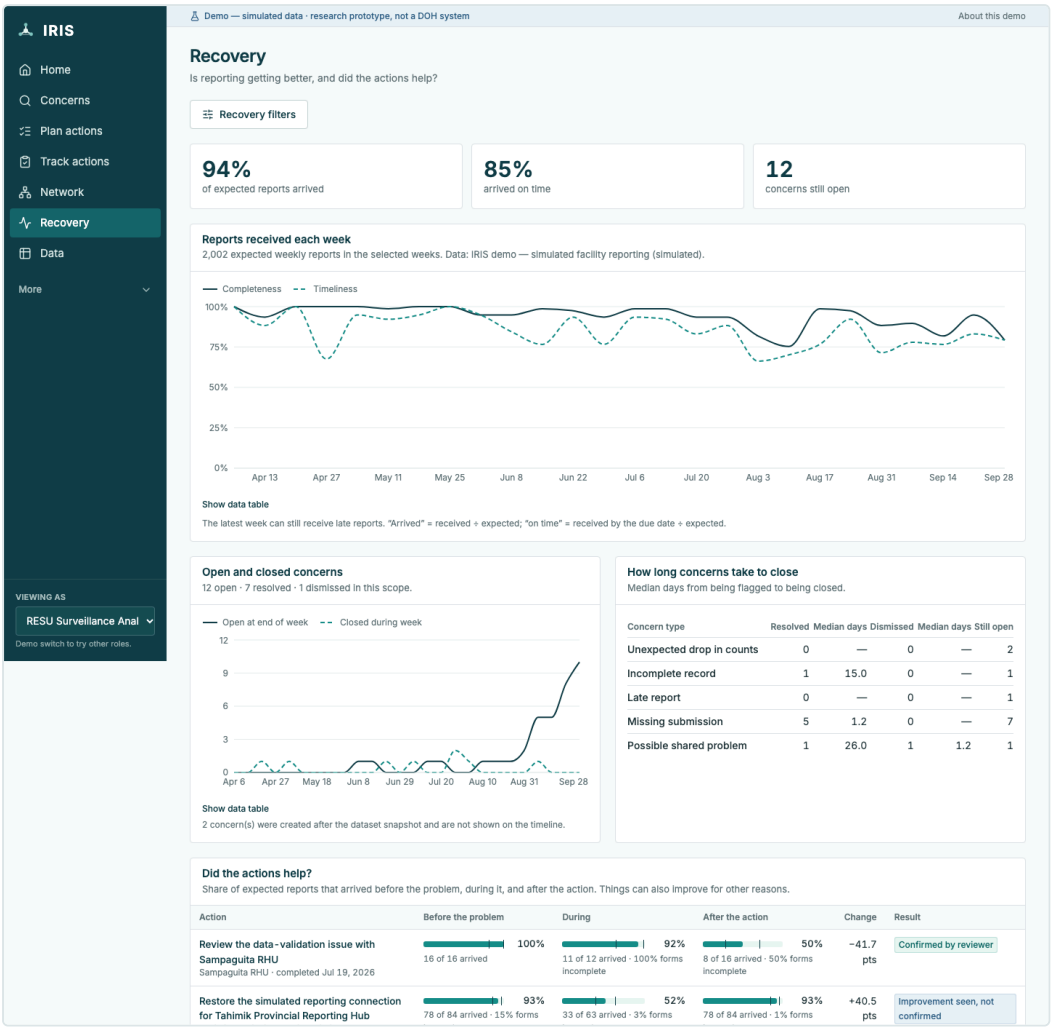


Figure 9. Three headline numbers, reports received each week, and how concerns were opened and closed.

- **Arrived** = reports received ÷ reports expected. **On time** = reports received by the due date ÷ reports expected.
- The chart shows both per week: solid line for arrived, dashed for on time. Select **Show data table** for the exact values.
- **Did the actions help?** compares reporting before the problem, during it, and after each completed action.
- If a dataset has no reporting schedule, IRIS says *“Not calculable from this dataset”* instead of guessing.

10. Data

The Data page lets you look at the records behind every concern.

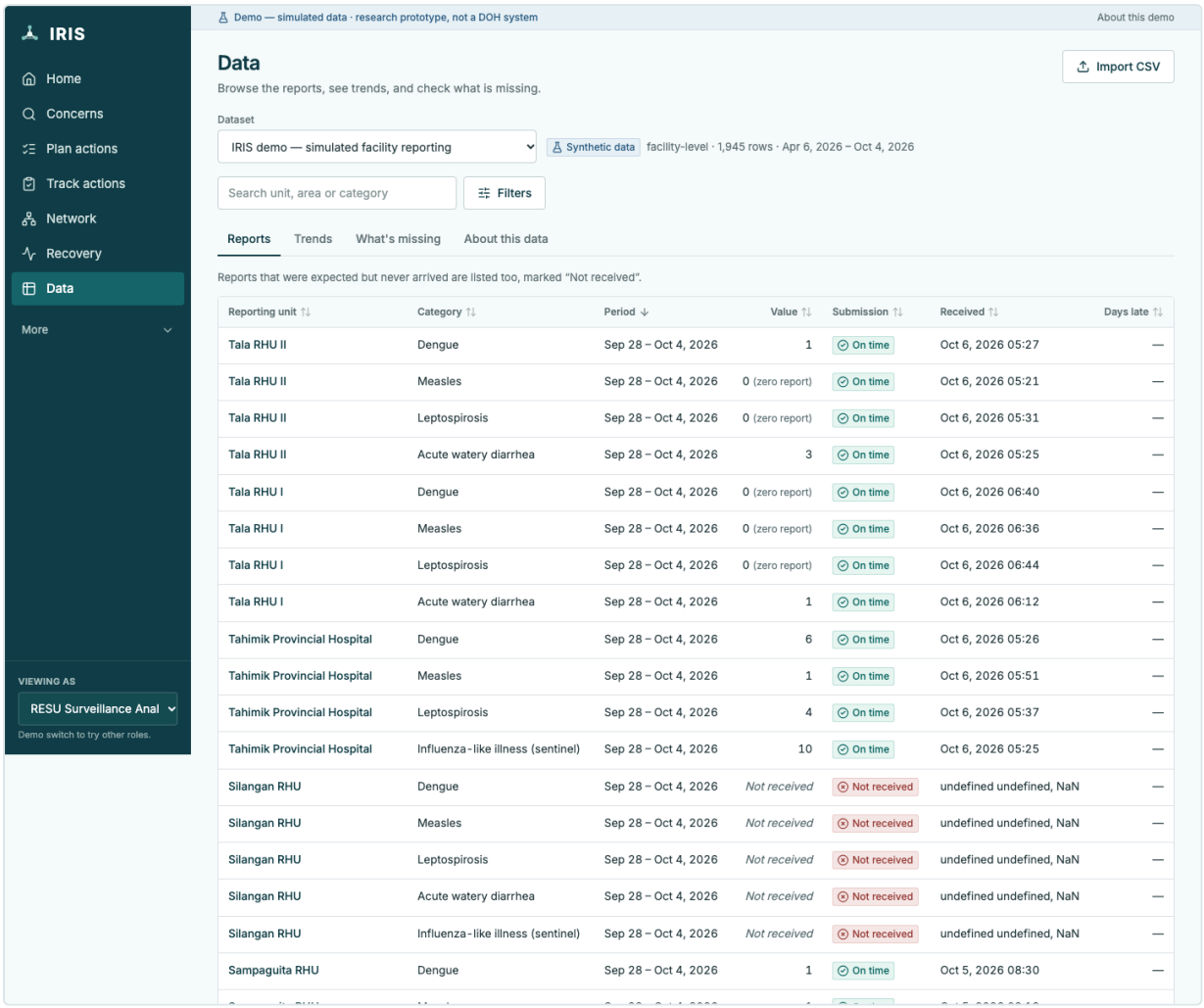


Figure 10. The Reports tab. Expected reports that never arrived are listed as "Not received", so gaps are visible.

The four tabs are **Reports** (select a row for its details and change history), **Trends** (one small chart per disease or province, on the same scale), **What's missing** and **About this data** (source, coverage, limitations and field definitions).

What's missing

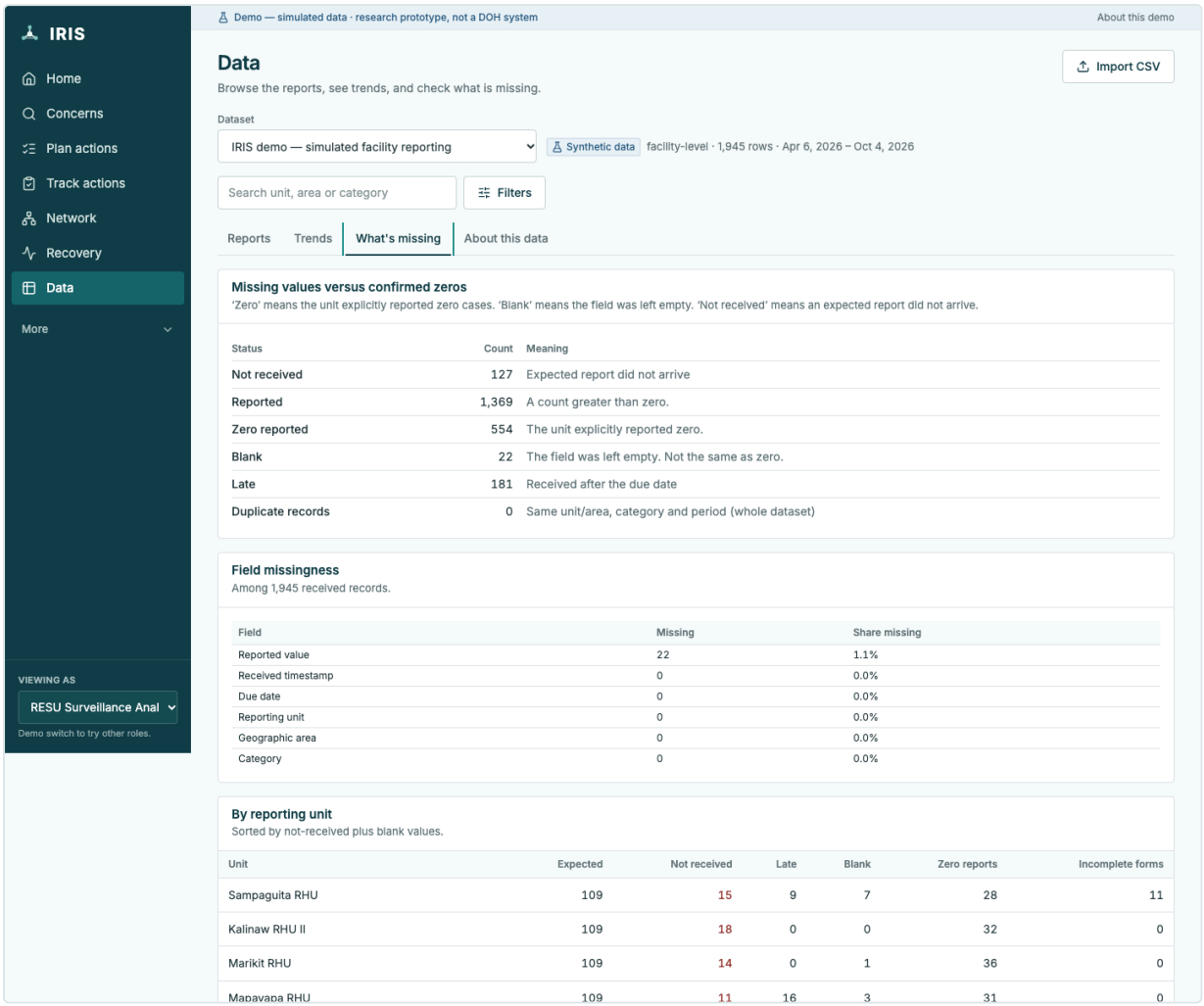


Figure 11. Missing values are kept separate from zeros, with a breakdown by reporting unit.

> **Blank is not zero.** *Zero reported* means the unit said there were no cases. *Blank* means the field was left empty. *Not received* means the report never arrived. IRIS never treats a blank or a missing report as zero.

11. Importing a CSV file

Analysts and administrators can add their own data from **Data** or **Data sources** using **Import CSV**. There are four short steps, and nothing is saved until you confirm.

1. **Upload** a CSV file (UTF-8, header row, up to 5 MB).
2. **Map columns**: tell IRIS which column is the week, the value, the disease, the area and so on. Likely matches are pre-selected.
3. **Validate**: IRIS lists every problem by row and column, such as unreadable dates, negative or non-numeric counts, unknown units or duplicate rows. Fix the file and upload again if needed.
4. **Describe and confirm**: name the dataset, give its source and say whether it is real, synthetic or of unknown origin.

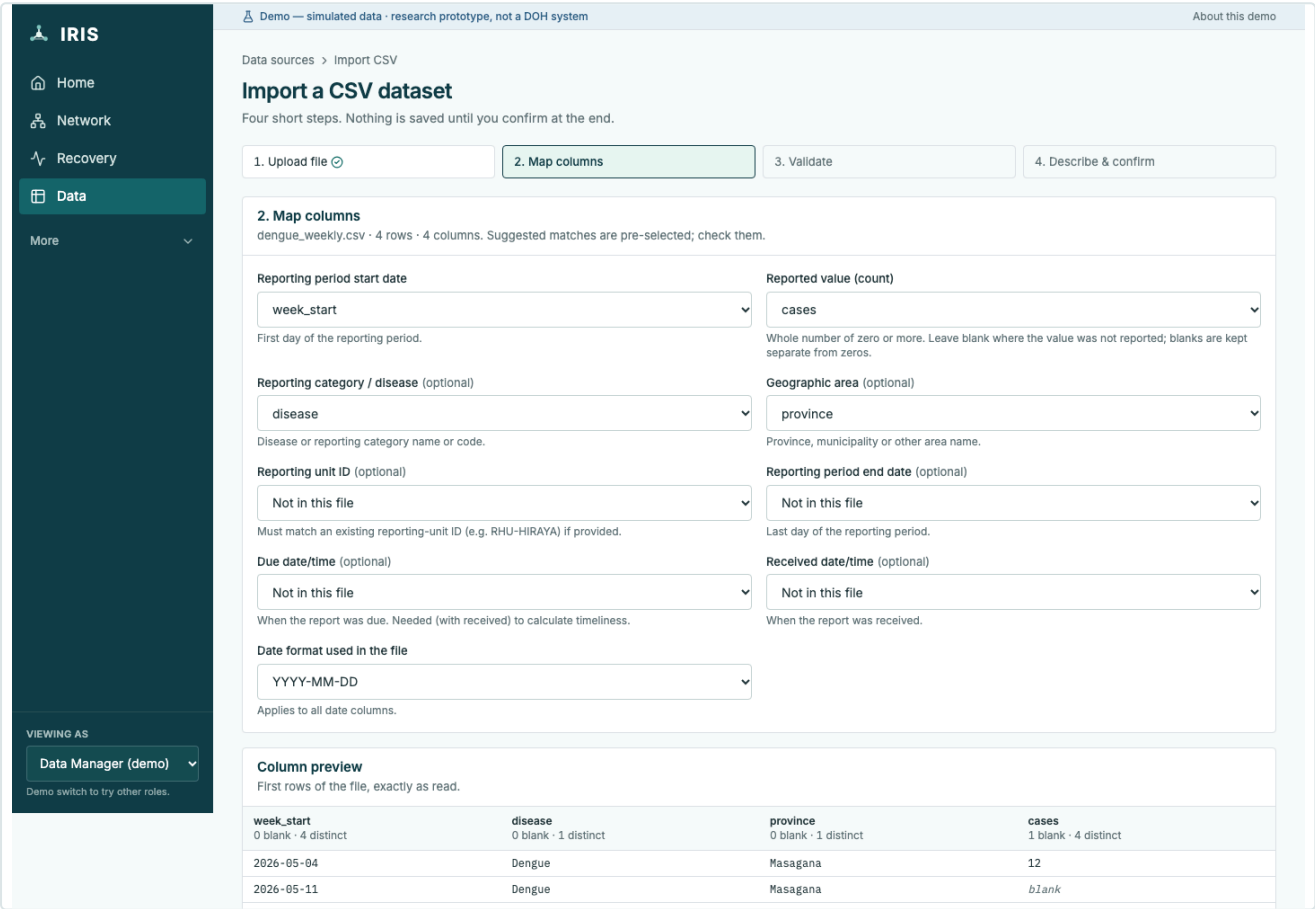


Figure 12. Step 2: mapping the file's columns, with a preview of the rows.

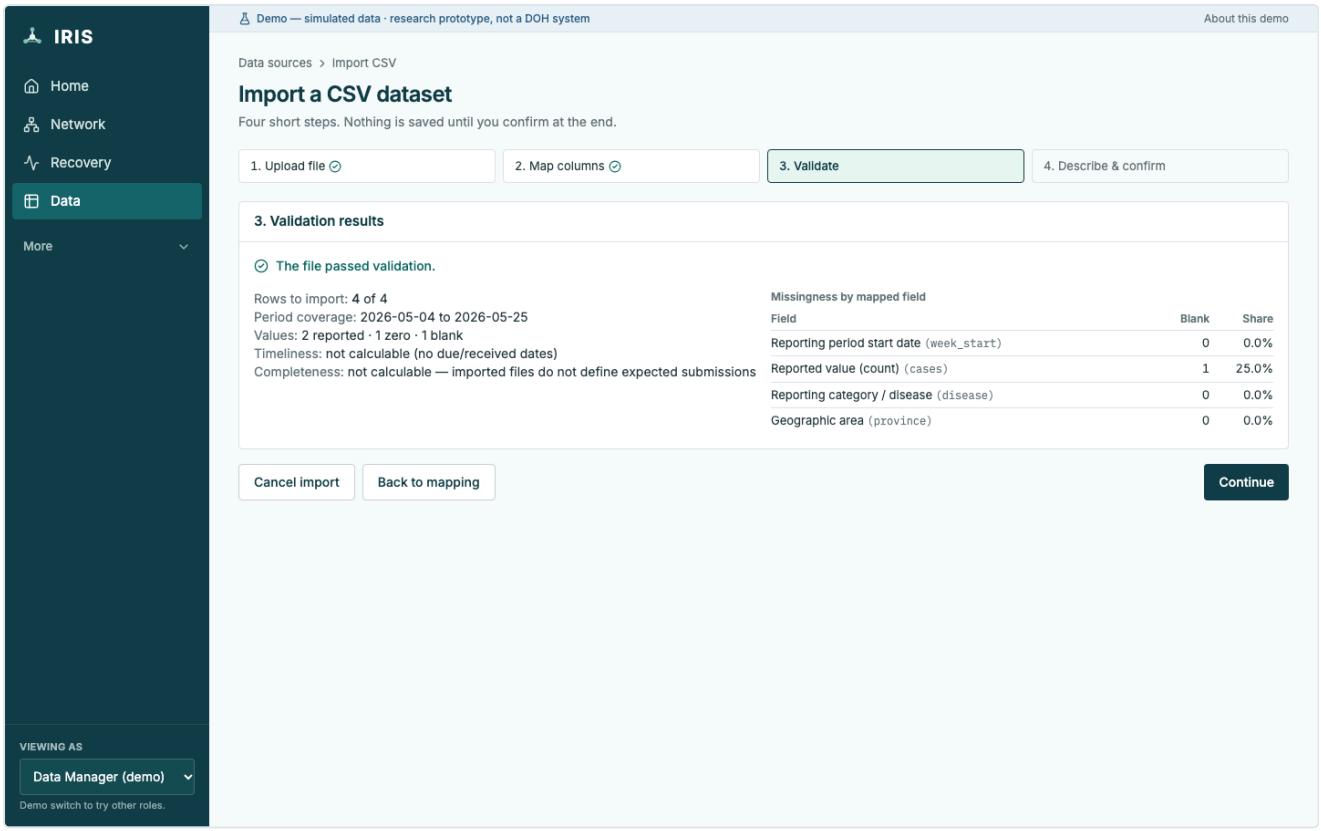


Figure 13. Step 3: the file passed validation. Note the counts of reported, zero and blank values.

12. Model evaluation

IRIS has three ways of spotting reporting problems. This page tests them on simulated data where the true answer is known, so you can see how often each is right.

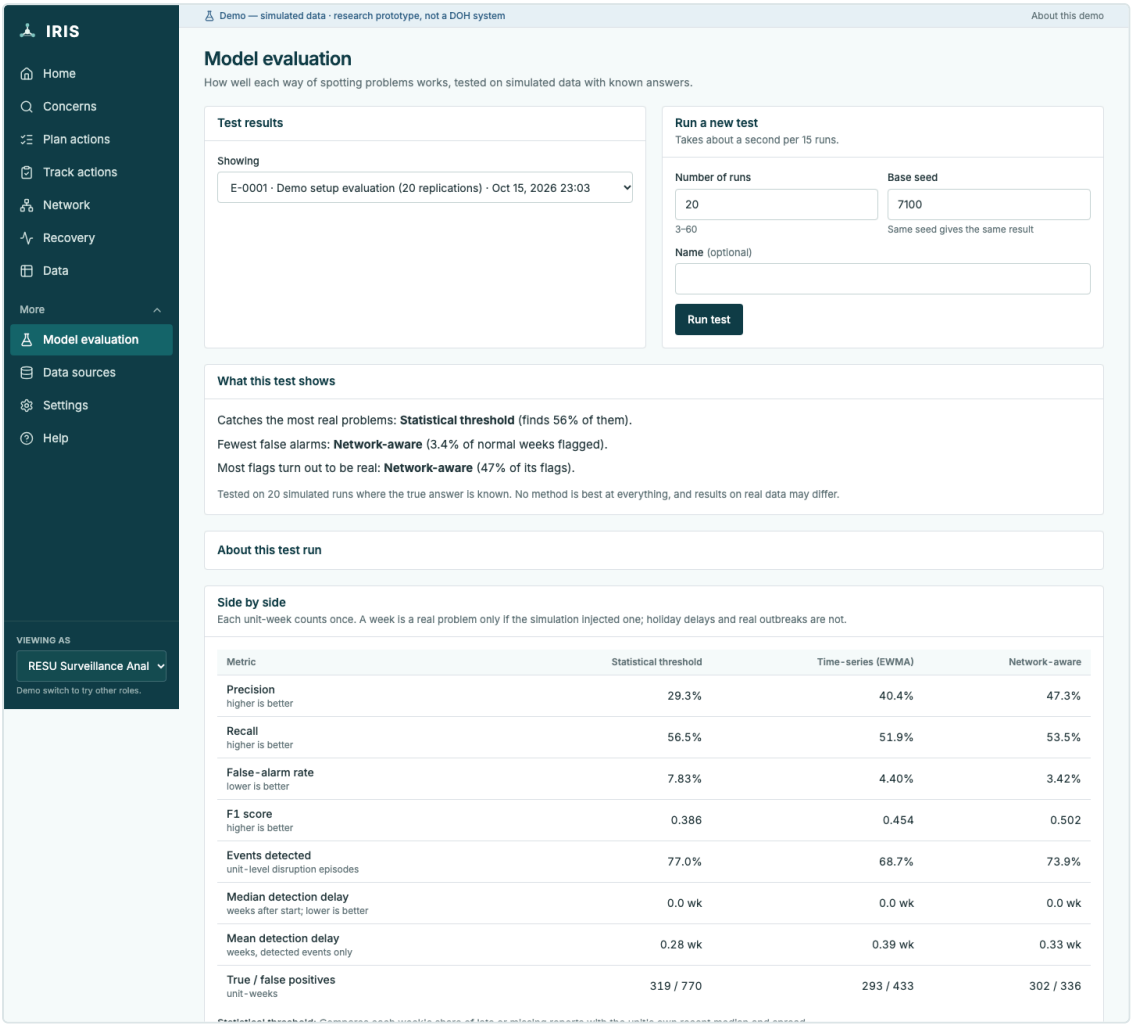


Figure 14. A plain summary, then the methods side by side.

Method	How it works
Statistical threshold	Compares each week with the unit's own recent weeks.
Time-series (EWMA)	Smooths the same signal over time to catch problems that last several weeks.
Network-aware	Also checks whether connected units had problems in the same week.

No method is best at everything: one may catch more problems while another raises fewer false alarms. Results are calculated from tests that were actually run, never typed in. Analysts can **Run a new test**, and **More results** breaks them down further.

13. Data sources and settings

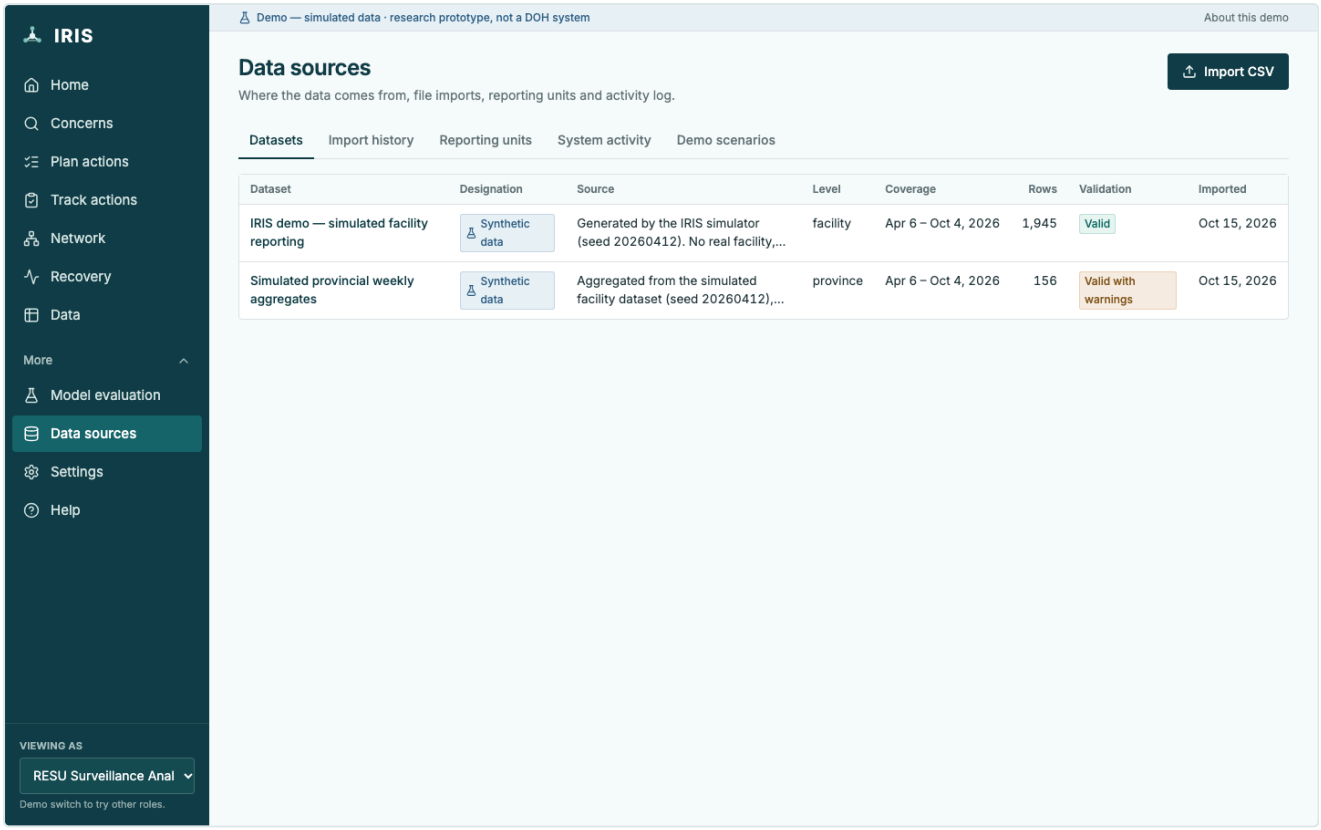


Figure 15. Data sources: each dataset with its origin, coverage and validation status.

Data sources (under More) has five tabs: **Datasets**, **Import history** (including rejected files and why), **Reporting units** (administrators can edit names, notes and status), **System activity** (every change, who made it and when) and **Demo scenarios** (choose which simulated problems to include and reset the demo).

Settings (under More) includes the role switch; table density and date format; larger text, reduced motion, underlined links and a stronger keyboard focus outline; the simulation answer key; and **Reset demo data**, which discards all demo changes.

14. Using IRIS on a phone

Every page works on a phone. The menu opens from the ≡ button, buttons are large enough to tap, and wide tables become cards or scroll on their own.

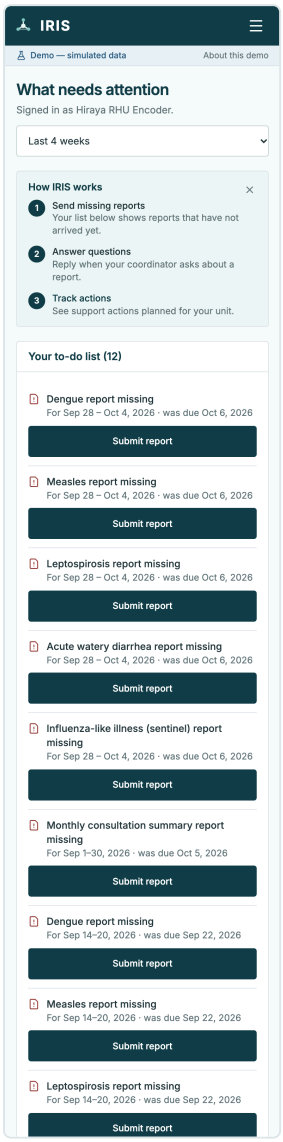


Figure 16. RHU staff to-do list.

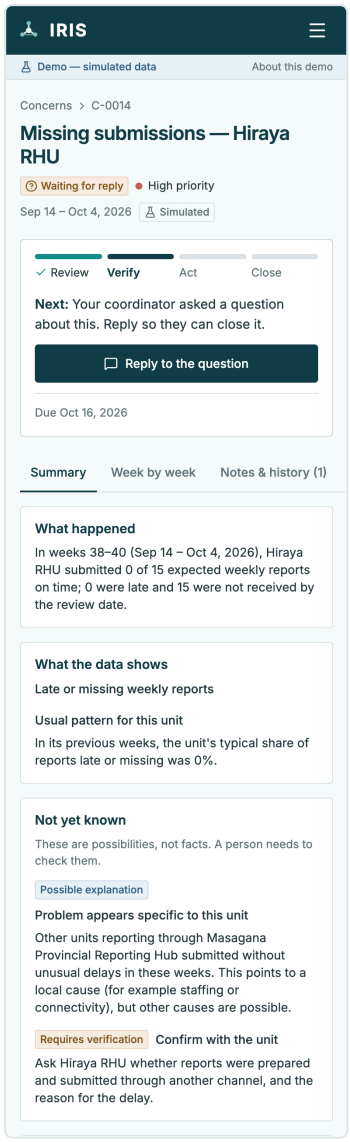


Figure 17. Replying to a coordinator's question.

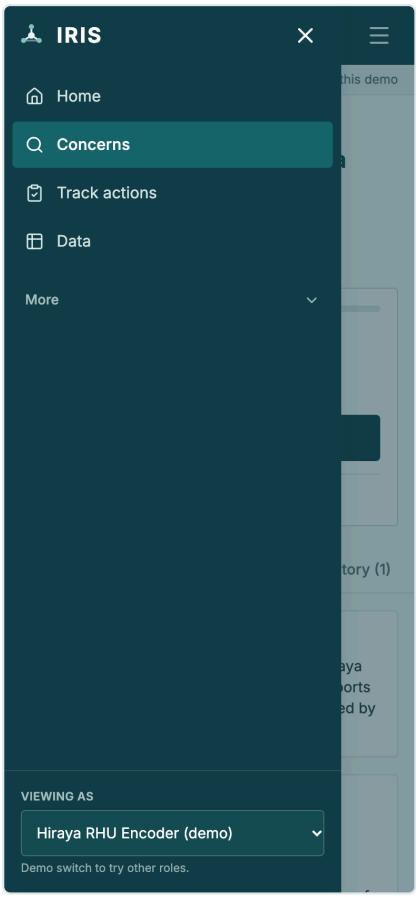


Figure 18. The menu for RHU staff.

For RHU staff

1. Open **Home**. *Your to-do list* shows reports that have not arrived.
2. Select **Submit report**, type the number of cases and submit. Enter **0** if there were no cases; tick *Value not available* only if the count really is unknown.
3. If your coordinator asked a question, select **Reply** and answer it.

15. Glossary

Term	Meaning
Reporting unit	A facility that sends reports: an RHU, hospital or laboratory.
Reporting hub	A point through which several units send their reports, such as a provincial office.
Expected reports	The reports a unit should send, according to its reporting schedule.
Concern	A possible reporting problem found in the data, waiting for a person to review it.
Possible shared problem	Connected units had problems at the same time; a common cause is possible but not proven.
Arrived (completeness)	Reports received ÷ reports expected.
On time (timeliness)	Reports received by the due date ÷ reports expected.
Staff-hours	The time available for corrective actions, used by Plan actions.
Catches problems (recall)	Of the real problems, the share a method found.
Flags that are real (precision)	Of the weeks a method flagged, the share that were real problems.
False alarm	A flag on a week with no real reporting problem.
Simulated	Generated by the IRIS simulator for demonstration; not real data.

16. Limitations

- IRIS is a research prototype, not an official or endorsed government system.
- All demonstration data, places, facilities and relationships are simulated. Province and municipality names are fictional.
- Detection settings and planning estimates were set for the simulation and have not been validated on real surveillance data.
- Model evaluation uses simulated scenarios with known answers; real-world performance may differ.
- “Did the actions help?” shows what happened afterwards, not proof that an action caused it.
- The role switch is for the demo only and is not real sign-in security.
- Data is stored in a local database file; files being imported are held in memory until confirmed and are lost if the server restarts.

Appendix A. Running IRIS on a computer

IRIS has two parts: a Python server that holds the data and calculations, and a web app that people use in the browser. You need Python 3.11 or newer with `uv`, and Node.js 20 or newer.

1. Get the code

```
git clone https://github.com/FerosC101/IRIS.git
cd IRIS
```

2. Start the server (terminal 1)

```
cd backend
uv sync
uv run uvicorn iris.main:app --port 8000
```

The first start creates the demo database and simulated data; this takes a few seconds.

3. Start the web app (terminal 2)

```
cd frontend
npm install
npm run dev
```

Open **http://localhost:5180** in a browser.

Checking that everything works

```
cd backend  && uv run pytest        # server tests
cd frontend && npm test             # web app tests
cd frontend && npm run qa:e2e       # clicks through the main workflows (servers running)
```

Updating this guide

With both parts running, `node scripts/capture-docs.mjs` (in `frontend`) retakes all screenshots, and `node scripts/build-user-guide.mjs` rebuilds this PDF from `docs/user-guide.html`.